Assignment-Regression Algorithm

Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on

the several parameters. The Client has provided the dataset of the same.

As a data scientist, you must develop a model which will predict the insurance

charges.

My Observations

As per the given Insurance data, insurance team maintained some parameter. And per client requirement as a Data scient we should predict the insurance charges based on the given parameter like Age, Sex, BMI, Children and Smoker

Data Set Design

1) 4 Inpit and 1 out column
2) This data set has 2 nominal columns (Sex , Smoker) we have convert them into numerical

Model & Stage Identification

1) Machine learning
2) Supervised learning
3) Regression

We can create the model based on the given data as follows

1) Muti linear
2) SVM
3) Decision Tree
4) Random Forest

   >> Based on the r2 value we can identify the best model

Multilinear

R score = **0.789479034986701**

SVM

S.No	Kernal	Gamma	C Value	R2_Score
1	Linear	Scale /Auto	0.1	**-0.080959968427891**
2	Linear	Scale /Auto	10	**0.462468414233968**
3	Linear	Scale /Auto	100	**0.6288792857320359**
4	Linear	Scale /Auto	1000	**0.7649311738608445**
5	Linear	Scale	10000	**0.7414230132360233**

6	Linear	Auto	10000	**0.7414230132360233**
7	rbf	Scale /Auto	0.1	**-.08907451521042731**
8	rbf	Scale /Auto	10	**-.03227329390671052**
9	rbf	Scale /Auto	100	**0.3200317832050831**
10	rbf	Scale /Auto	1000	**0.8102064851758545**
11	rbf	Scale /Auto	10000	**0.8779952401449916**
12	poly	Scale /Auto	100	**0.6179569624059795**
13	Poly	Scale /Auto	1000	**0.8566487675946551**
14	poly	Scale /Auto	10000	**0.8591715095262809**
15	sigmoid	Scale /Auto	100	**0.5276103546510407**
16	sigmoid	Scale /Auto	1000	**0.28747069486976695**
17	sigmoid	Scale /Auto	10000	**-34.151535978496256**

Decision Tree

S.No	**Kernal**	**Spliter**	**Max Depth**	**R2_Score**
1	Squared_error	Best	2	**0.8569635601474321**
2	Squared_error	Best	3	**0.8751026719462079**
3	Squared_error	Best	10	**0.7375712347796017**
4	Squared_error	Random	2	**0.7041115754202941**
5	Squared_error	Random	3	**0.7575845108606476**
6	Squared_error	Random	100	**0.7124388330253346**
7	Friedman_mse	Best	3	**0.8751026719462079**
8	Friedman_mse	Best	**10**	**0.7240704353591292**
9	Friedman_mse	Best	**100**	**0.6867103434196301**
10	Friedman_mse	random	**3**	**0.7936302813346694**
11	Friedman_mse	random	**10**	**0.8433387543865267**
12	Friedman_mse	random	**15**	**0.7218511913587462**
13	Absolute_error	Best	**2**	**0.8513874442378325**
14	Absolute_error	Best	**3**	**0.8679279967169236**

15	Absolute_error	Best	**10**	**0.7562741378648761**
16	Absolute_error	random	**3**	**0.770004914724915**
17	Absolute_error	random	**5**	**0.8792169198718444**
18	Absolute_error	random	**10**	**0.7944572706835437**
19	Poisson	Best	**2**	**0.8569635601474321**
20	Poisson	Best	**3**	**0.8778038152473475**
21	Poisson	Best	**10**	**0.7371714157688705**
22	Poisson	random	**2**	**0.7184451639595265**
23	Poisson	random	**3**	**0.8027484554650799**
24	Poisson	random	**10**	**0.8048049564556428**

4. Random Forest

S.No	**Criterion**	**N_estimator**	**Random_State**	**R2_Score**
1	Squared_error	**50**	**0**	**0.8498329315421834**
2	Squared_error	**100**	**0**	**0.8538307913484513**
3	Squared_error	**200**	**0**	**0.8524777475124578**
4	absolute_error	**10**	**0**	**0.835063555313752**
5	absolute_error	**50**		**0.8526655993519747**
6	absolute_error	**100**	**0**	**0.8520093621081837**
7	Friedman_mse	**10**	**0**	**0.8520093621081837**
8	Friedman_mse	**50**	**0**	**0.8500716139332296**
9	Friedman_mse	**100**	**0**	**0.8540518935149612**
10	poisson	**50**	**0**	**0.8491075958392151**

11	poisson	**100**	**0**	**0.8526334258892607**
12	poisson	**200**	**0**	**0.8523005268627799**

Result: As per the above all regression execution with different parameters we can Go with Decision Tree is the best model. Because the R2 values is **0.8792169198718444 .**

Even though SVM regression R2 values are high **0.8779952401449916** C value (100) parameter value is High